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25 August 1976
SUPERSEDING
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MILITARY SPECIFICATION

DELAY LINES, PULSE, ELECTROMAGNETIC, FIXED

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

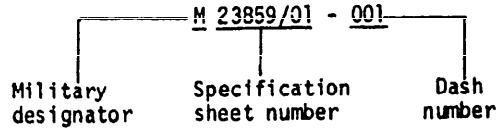
1. SCOPE

1.1 Scope. This specification covers the general requirements for fixed (tapped, untapped and sectionalized) electromagnetic pulse delay lines for use in Armed Services electronic equipment. This specification is not intended to cover active lines, delay cables, electro-mechanical, magnetostrictive, photoelastic, semi-conductive, or acoustic delay devices.

1.2 Classification.

1.2 Military part number. The military part number shall consist of the letter "M", basic number of the specification sheet, and an assigned dash number as shown in the following example (see 3.1 and 6.2.1).

EXAMPLE:



2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specifcation to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571 - Solder, Tin Alloy: Tin-Lead Alloy; and Lead Alloy.

MILITARY

MIL-F-14256 - Flux, Soldering, Liquid (Rosin Base).
MIL-E-17555 - Electronic and Electrical Equipment, Accessories and Repair Parts: Packaging and Packing of.
MIL-C-45662 - Calibration System Requirements.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking Of U.S. Military Property.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

ELECTRONIC INDUSTRIES ASSOCIATION STANDARD

RS-242 - Definitions for Electro-magnetic Delay Lines.

(Application for copies should be addressed to the Electronic Industries Association Engineering Office, 11 West 42nd Street, New York, N.Y. 10036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the individual specification sheets, the latter shall govern. (See 6.2).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.3).

3.3 Definitions and symbols. Definitions and symbols shall be in accordance with EIA Standard RS-242.

3.4 Materials. The materials shall be as specified herein; however, when a definite material is not specified, a material shall be used which will enable the delay line to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Flammable materials. Insofar as practicable, materials used in the construction of delay lines shall be nonflammable and nonexplosive.

3.4.2 Corrosive materials. Corrosive materials used in any of the manufacturing processes shall be removed or neutralized so that no corrosion will result from such use. Insofar as practicable, materials used in the construction of delay lines shall be noncorrosive.

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3.4.3 Solder and soldering flux. Solder, when used, shall be in accordance with QQ-S-571. Soldering flux shall be in accordance with MIL-F-14256.

3.4.4 Screws, nuts and washers. All mounting and terminal screws, nuts, and washers shall be of corrosion-resistant material or shall be protected against corrosion.

3.5 Design and construction. Delay lines shall be of the design, construction, and physical dimensions as specified in the applicable specification sheet (see 3.1).

3.5.1 Mounting screws and mounting inserts. Screw threads shall be class 2A or 2B, as applicable (see 3.1), in accordance with Handbook H28. External screw threads, class 2 fit, shall, after receiving a finish, be capable of accepting a nut of class 2B fit. Internal screw threads, class 2 fit, shall, after receiving a finish, be capable of accepting a screw of class 2A fit. Maximum external and internal installation torque shall be in accordance with the following:

Screw size	Torque (pound-inches)
.112-40 UNC - - -	3
.138-32 UNC - - -	5
.164-32 UNC - - -	6
.190-32 UNF - - -	8
.250-20 UNC - - -	8
.3125-18UNC - - -	8

Nuts shall run down to within two threads of mounting surface.

3.5.2 Case material. Unless otherwise specified (see 3.1), cases may be of metallic or nonmetallic material. All metallic surfaces shall be suitably protected against corrosion by a suitable finish and shall be free from blisters and other defects which may affect the protective value of this finish.

3.5.3 Terminals. Terminals shall be solder lug terminals, printed-circuit and dual-in-line terminals, or solid-wire lead terminals, as specified (see 3.1). Unless otherwise specified, solder lug terminals shall be capable of accommodating two No. 22 AWG wires. All solder type terminals shall be capable of complying with the solderability requirements of this specification.

3.6 Solderability. When delay lines are tested as specified in 4.6.1, they shall meet the applicable criteria for terminal evaluation in the test method.

3.7 Resistance to solvents. When delay lines are tested as specified in 4.6.2, there shall be no evidence of mechanical damage and the markings shall remain legible. The paint or exterior finish shall not soften, peel or show other signs of deterioration.

3.8 Resistance to soldering heat. When delay lines are tested as specified in 4.6.4, there shall be no softening of the insulation or loosening of the windings or terminals.

3.9 Terminal strength. When delay lines are tested as specified in 4.6.5.1 to 4.6.5.2 inclusive, there shall be no evidence of loosening, rupturing, or other mechanical damage. Bends shall not be considered as damage unless surface cracking is evident. Except for flexible leads, there shall be no rotation of the terminals. Rotation of the external portion of the metallic portion of a "hook" type terminal exceeding 10 degrees shall not constitute a failure.

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3.10 Seal. When delay lines are tested as specified in 4.6.6 there shall be no evidence of continuous air bubble flow or compound leakage.

3.11 Dielectric withstanding voltage. When delay lines are tested as specified in 4.6.7, there shall be no arcing, flashover, breakdown of insulation, or other evidence of damage.

3.12 Insulation resistance (when applicable). When delay lines are tested in accordance with 4.6.8, the minimum insulation resistance shall be 1000 megohms, unless otherwise specified (see 3.1).

3.13 Electrical characteristics. Delay lines shall be capable of meeting all the electrical requirements specified herein.

3.13.1 Delay time. Unless otherwise specified, when tested in accordance with 4.6.9.1.1, the overall specified delay time of the line shall be within ±5 percent or ±1 nanosecond, whichever is greater (see 3.1). In the case of sectionalized delay lines, the tandem combination of all delay sections and any combination of sections shall be within ±5 percent or ±1 nanosecond, whichever is greater, of the delay time specified (see 3.1).

3.13.2 Rise time. When measured in accordance with 4.6.9.1.2, the rise time of pulses taken at the point of maximum delay shall not exceed the quotient of the measured delay time divided by the applicable delay to rise-time ratio. The rise time measured at the taps or for sections of sectionalized delay lines shall not exceed the rise time required at the point of maximum delay.

3.13.3 Attenuation. When measured in accordance with 4.6.9.1.3, the attenuation of pulses taken at the point of maximum delay with respect to the input pulse amplitude shall not exceed the value specified (see 3.1). The attenuation measured at the taps or for sections of sectionalized delay lines shall not exceed the attenuation required at the point of maximum delay.

3.13.4 Distortion. Unless otherwise specified (see 3.1), when measured in accordance with 4.6.9.1.4, the all encompassing distortion of pulses appearing at the point of maximum delay shall not exceed ±10 percent. The distortion measured at the taps or for sections of sectionalized delay lines shall not exceed the distortion required at the point of maximum delay.

3.13.5 Delay time variation with temperature. When tested in accordance with 4.6.9.1.5, the delay time variation with temperature shall not exceed the value specified (see 3.1).

3.13.6 Thermal-age stability. When tested in accordance with 4.6.9.1.6, the change in the electrical measurements recorded between readings prior to environmental tests, and readings after environmental tests, shall not exceed the values specified (see 3.1).

3.13.7 Figure of merit. The figure of merit shall be as specified in the applicable specification sheet.

3.13.8 Nominal terminating impedance. The terminating impedance of the delay line shall be resistive and within ±1 percent of the value specified in the applicable specification sheet.

3.14 Salt spray (corrosion) (when specified, see 3.1). When delay lines are tested as specified in 4.6.10, there shall be no evidence of corrosion as exhibited by any visible degradation of the surfaces that can be attributed to flaking, pitting, blistering or otherwise loosened protective coating or metal surface.

3.15 Vibration. When delay lines are tested as specified in 4.6.11, there shall be no leakage of filling material, no evidence of other physical damage such as cracks, bursting, or bulging of the case. There shall be no evidence of mechanical damage and there shall be no electrical discontinuity during the test.

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3.16 Shock. When delay lines are tested as specified in 4.6.12, there shall be no leakage of filling material, no evidence of other physical damage such as cracks, bursting, or bulging of the case. There shall be no evidence of mechanical damage and there shall be no electrical discontinuity during the test.

3.17 Winding continuity. When delay lines are tested as specified in 4.6.13, all windings shall be electrically continuous.

3.18 Thermal shock. When delay lines are tested as specified in 4.6.14, not more than 10 percent of the surface shall have peeling, flaking, chipping, cracking, or other impairment of the protective finish; no evidence of other physical damage such as cracks, bursting, or bulging of the case; or other defects that would affect the mechanical or electrical operation.

3.19 Immersion. When tested in accordance with 4.6.15, delay lines shall show no evidence of continuous air bubble flow or compound leakage.

3.20 Moisture resistance. When tested in accordance with 4.6.16, there shall be no evidence of other physical damage that would affect the mechanical or electrical operation of the line.

3.21 Flammability (when specified, see 3.1). When delay lines are tested as specified in 4.6.17, there shall be no evidence of violent burning which results in an explosive-type flame, and the delay lines subjected to this test shall be self extinguishing.

3.22 Life (when specified, see 3.1). When delay lines are tested as specified in 4.6.18, there shall be no evidence of impairment to the protective finish or of other physical damage such as cracks, bursting, or bulging of the case.

3.23 Fungus. The manufacturer shall certify that all external materials are non-nutrient to fungus growth or are suitably treated to retard fungus growth, or shall perform the test specified in 4.6.19. When delay lines are tested as specified in 4.6.19, there shall be no evidence of fungus growth on the external surface.

3.24 Marking. Delay lines shall be marked in accordance with MIL-STD-130, with the military part number, manufacturer's part number, source code, terminal and tap identification, and date code. The date code shall be in accordance with MIL-STD-1285. The marking may be applied to more than one side of the case if the required marking requires more space than is available on the one side. Where the surface areas are insufficient for all the required information, as many as possible of the markings shall be applied using the order of priority as listed above. Markings shall remain legible after all tests.

3.24.1 Terminal and tap identification. Unless otherwise specified (see 3.1) terminals shall be identified as follows:

Input - - - - - - - - - "IN"

Ground- - - - - - - - - "GRD"

Taps including output terminal shall be marked sequentially beginning with numeral 1 following the input (IN) terminal.

3.25 Workmanship. Delay lines shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy, quality and quantity to permit performance of the required inspection shall be established and maintained by the supplier. The establishment and maintenance of a calibration system to control the accuracy of the measuring and test equipment shall be in accordance with MIL-C-45662.

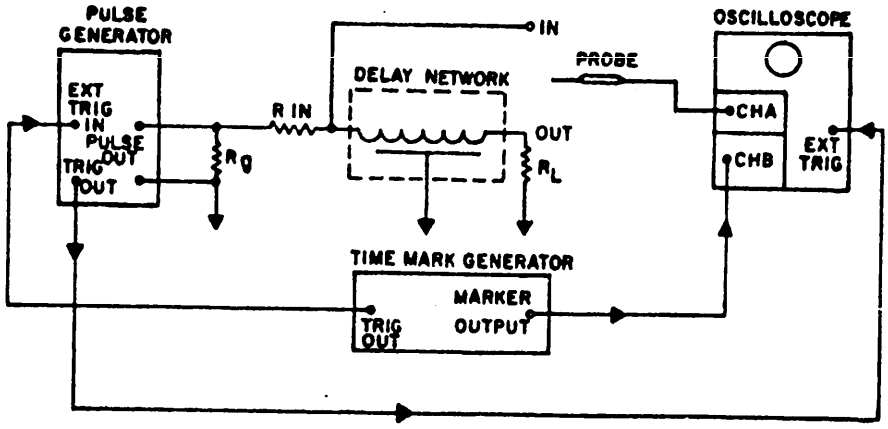
4.2 Classification of inspections. The inspections specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202.

4.3.1 Standard test circuit, test equipment and measurements. Standard test circuit, test equipment and measurements shall be as specified in 4.3.1.1 through 4.3.1.6.

4.3.1.1 Standard test circuit. The standard test circuit shall be as shown on figure 1. The pulse generator shall be loaded with an L pad to match delay line impedance to generator. The delay line shall be terminated in the nominal terminating impedance within ±1 percent. The test circuit capacitances including probe shall be kept to a minimum.



NOTE: All measurements refer to definitions of EIA Standard RS-242.
R in = Input matching resistance.
Rg = Generator terminating impedance.
R_L = Terminating resistance.

FIGURE 1. Standard test circuit.

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4.3.1.2 Test equipment. The test equipment shall be as follows:

(a) Marker generator - Tektronix type 180 or equivalent.
(b) Pulse generator - Hewlett Packard type 215A or E-H Model 139B, or equivalent. Type 214B or 214A, or equivalents, may be used if rise time of line being tested is greater than 0.06 µs.
(c) Oscilloscope - Tektronix type 545 or equivalent for delay measurements and for pulse rise time measurements exceeding 0.06 µs. Tektronix type 561 or equivalent for pulse rise time measurements 0.06 µs, or less.

4.3.1.3 Delay measurements. The basic steps for measurement of delay shall be as follows:

(a) Establish reference $t_d = 0$.
(b) Sweep speed: Use sweep speed such that the output rise is displayed across 4-5 cm so that it crosses the screen at approximately 45 degrees. The markers may then be chosen so as to provide a mark each 1-5 cm.
(c) Display 0.1 µs markers on channel B (or 10 MHz sinewave).
(d) Display input pulse on channel A. Set again for 4.0 cm deflection E_{in}.
(e) Adjust "pulse position vernier" on pulse generator for coincidence of 50 percent amplitude point of input pulse and a reference marker.
(f) To measure delay of output pulse move channel A probe to output test point.
(g) Rotate "delay time multiplier" on oscilloscope, counting markers, until the output pulse is displayed.
(h) Adjust gain on channel A to provide 4.0 cm deflection of output pulse (E_0).
(i) Delay equals the number of 0.1 µs markers plus increment between last marker and 50 percent point of output pulse.

4.3.1.4 Rise time measurements. Rise time measurements shall be made as follows:

(a) Set sweep speed such that the output rise is displayed across 4-5 cm so that it crosses the screen at approximately 45 degrees. The markers may then be chosen so as to provide a mark each 1-5 cm.
(b) Display 0.1 µs markers or shorter markers if available.
(c) Display input pulse (input test point) on channel A. Set gain for 4.0 cm deflection (E_1).
(d) Measure input pulse rise time (t_{ri}).
(e) To measure output rise time move channel A probe to output test point.
(f) Rotate "delay time multiplier" on oscilloscope until the output pulse is displayed.
(g) Adjust gain of channel A to provide 4.0 cm deflection of output pulse (E_0).
(h) Measure output pulse rise time (t_{ro}).

4.3.1.5 Distortion measurements. Distortion measurements shall be made as follows:

(a) Adjust scope same as for delay time measurement.
(b) Display output pulse.
(c) Measure amplitudes of output pulse distortion b, c, and d.
(d) Calculate delay line output pulse distortion (S_0).

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4.3.1.6 Attenuation measurements. Attenuation measurements shall be made as follows:

(a) Connect a short between the "in" and "out" points shown in figure 1.
(b) Adjust scope to display input pulse amplitude as for delay time.
(c) Remove short and replace with the delay line under test.
(d) Move probe to output test point.
(e) Rotate "delay time multiplier" on oscilloscope to display output pulse.
(f) Measure output pulse amplitude.
(g) Calculate attenuation.

4.4 First article inspection. First article inspection shall be performed by the contractor, after award of contract and prior to production, at a location acceptable to the Government. First article inspection shall be performed on sample units which have been produced with equipment and procedures normally used in production. First article approval is valid only on the contract under which it is granted, unless extended by the Government to other contracts.

4.4.1 Sample size. Eight sample units shall be subjected to first article inspection; however, two additional sample units shall be required if certification is not provided for the fungus test. Only six sample units shall be required when the life test is not performed.

4.4.2 Inspection routine. The sample units shall be subjected to the inspections specified in table I, in the order shown, and to the tests specified in the individual specification sheet (see 3.1).

TABLE I. First article inspection.

Examination or test	Requirement paragraph	Method paragraph
Group I (all sample units)		
Solderability 1/	3.6	4.6.1
Resistance to solvents (3 sample units) 2/	3.7	4.6.2
Visual and mechanical examination (external)	3.1,3.4,3.5, 3.24 and 3.25	4.6.3
Resistance to soldering heat 1/	3.8	4.6.4
Terminal strength	3.9	4.6.5
Seal	3.10	4.6.6
Dielectric withstanding voltage (at atmospheric pressure)	3.11	4.6.7.1
(at barometric pressure) (when specified)	3.11	4.6.7.2
Insulation resistance	3.12	4.6.8
Electrical characteristics		
Delay time	3.13.1	4.6.9.1.1
Rise time	3.13.2	4.6.9.1.2
Attenuation	3.13.3	4.6.9.1.3
Distortion	3.13.4	4.6.9.1.4
Delay time variation with temperature	3.13.5	4.6.9.1.5
Thermal-age stability	3.13.6	4.6.9.1.6

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TABLE I. First article inspection. -Continued

Examination or test	Requirement paragraph	Method paragraph
Group II (6 sample units)		
Salt spray (when specified)	3.14	4.6.10
Vibration	3.15	4.6.11
Shock	3.16	4.6.12
Winding continuity	3.17	4.6.13
Dielectric withstanding voltage (reduced voltage)	3.11	4.6.7
Thermal shock	3.18	4.6.14
Winding continuity	3.17	4.6.13
Dielectric withstanding voltage (reduced voltage)	3.11	4.6.7
Insulation resistance	3.12	4.6.8
Immersion	3.19	4.6.15
Moisture resistance	3.20	4.6.16
Insulation resistance	3.12	4.6.8
Dielectric withstanding voltage (reduced voltage)	3.11	4.6.7
Electrical characteristics		
Delay time	3.13.1	4.6.9.1.1
Rise time	3.13.2	4.6.9.1.2
Attenuation	3.13.3	4.6.9.1.3
Distortion	3.13.4	4.6.9.1.4
Thermal-age stability	3.13.6	4.6.9.1.6
Visual and mechanical examination (external)	3.1,3.4,3.5, 3.24, and 3.25	4.6.3
Flammability (2 sample units) (when specified)	3.21	4.6.17
Group III (2 sample units)		
Life (when specified)	3.22	4.6.18
Dielectric withstanding voltage: (At reduced voltage)	3.11	4.6.7.3
Insulation resistance	3.12	4.6.8
Electrical characteristics (as specified)	3.13	4.6.9
Visual and mechanical examination (external)	3.1,3.4,3.5, 3.24 and 3.28	4.6.3
Group IV		
Fungus 3/	3.23	4.6.19

1/ Solderable type terminals only.
2/ Printed circuit type delay lines only or when specified (see 3.1).
3/ Test shall not be performed if the manufacturer provides certification that all external materials are non-nutrient to fungus growth or suitably treated to retard fungus growth.

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4.4.3 Failures. One or more failures shall be cause for refusal to grant first article approval.

4.5 Quality conformance inspection.

4.5.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A and B inspection.

4.5.1.1 Inspection lot. An inspection lot shall consist of all delay lines of the same type designation, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table II, in the order shown.

4.5.1.2.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table II. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.2.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

TABLE II. Group A inspection.

Examination or test	Requirement paragraph	Test method paragraph	AQL (percent defective)	
			Major	Minor
Visual and mechanical examination (external)- - - - - - - - - - -	3.1,3.4,3.5, 3.24 and 3.25	4.6.3	1.0	4.0
Seal - - - - - - - - - - - - - - -	3.10	4.6.6	1.0	---
Dielectric withstanding voltage (at atmospheric pressure) - - -	3.11	4.6.7.1	1.0	---
Dielectric withstanding voltage (at reduced barometric pressure)(when applicable)- - -	3.11	4.6.7.2		
Insulation resistance- - - - - -	3.12	4.6.8		

4.5.1.3 Group B inspection. Group B inspection shall consist of the test specified in table III and shall be performed on sample units that have been subjected to and have passed the group A inspection unless it is more practical to select a separate sample from the lot for group B inspection.

4.5.1.3.1 Sampling plan. The sampling plan shall be in accordance with MIL-STD-105 for special inspection level S4. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The sample size shall be based on the inspection lot size from which the sample was selected for group A inspection. The AQL shall be 6.5 percent defective.

TABLE III. Group B inspection.

Test	Requirement paragraph	Method paragraph
Electrical characteristics (as applicable)- - -	3.13	4.6.9

4.5.1.3.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

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4.5.1.3.3 Disposition of sample units. Sample units which have passed all the group B inspection shall be delivered on the contract or order, if the lot is accepted.

4.5.2 Periodic check test. Periodic check test shall consist of group C. Except where the results of these tests show noncompliance with the applicable requirements (see 4.5.2.1.4), delivery of products which have passed groups A and B shall not be delayed pending the results of group C inspection.

4.5.2.1 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table IV, in the order shown. Group C inspection shall be made on sample units selected from inspection lots which have passed the groups A and B inspection.

TABLE IV. Group C inspection.

Examination or test	Requirement paragraph	Method paragraph
Subgroup I		
Resistance to soldering heat	3.8	4.6.4
Terminal strength 2/	3.9	4.6.5
Salt spray (when applicable)	3.14	4.6.10
Vibration	3.15	4.6.11
Shock	3.16	4.6.12
Winding continuity	3.17	4.6.13
Dielectric withstanding voltage (reduced voltage)	3.11	4.6.7
Thermal shock	3.18	4.6.14
Winding continuity	3.17	4.6.13
Dielectric withstanding voltage	3.11	4.6.7
Insulation resistance	3.12	4.6.8
Immersion	3.19	4.6.15
Moisture resistance	3.20	4.6.16
Insulation resistance	3.12	4.6.8
Dielectric withstanding voltage (reduced voltage)	3.11	4.6.7.3
Electrical characteristics		
Delay time	3.13.1	4.6.9.1.1
Rise time	3.13.2	4.6.9.1.2
Attenuation	3.13.3	4.6.9.1.3
Distortion	3.13.4	4.6.9.1.4
Visual and mechanical examination (external)	3.1,3.4,3.5, 3.24 and 3.25	4.6.3
Resistance to solvents (3 sample units) 1/	3.7	4.6.2
Flammability (2 sample units)(when specified)	3.21	4.6.17
Subgroup 2		
Solderability 2/	3.6	4.6.1
Life (when specified)	3.22	4.6.18
Dielectric withstanding voltage: (At reduced voltage)	3.11	4.6.7.3
Insulation resistance	3.12	4.6.8
Electrical characteristics (as specified)	3.13	4.6.9
Visual and mechanical examination (external)	3.1,3.4,3.5, 3.24 and 3.25	4.6.3

1/ Printed-circuit type terminals only.
2/ Once a particular type of terminal has been subjected to the solderability and terminal strength tests, other delay lines containing that terminal type need not be subjected to these tests for the remainder of the 12-month period from which that terminal type was selected.

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4.5.2.1.1 Sampling plan.

4.5.2.1.1.1 Subgroup 1. Every 12 months, 4 sample units of any impedance value of each delay line style produced during the month prior to the annual inspection, shall be subjected to the inspection of group I. If, during a 12-month period, first article testing has been successfully completed on delay lines similar in basic mechanical design as defined in 4.5.2.1.2, subgroup 1 inspection may be omitted for that 12-month period.

4.5.2.1.1.2 Subgroup 2. Every 12 months, 2 sample units of any impedance value of each delay line style shall be subjected to the tests in subgroup 2, in the order shown. If during the 12-month period, first article testing has been successfully completed on delay lines similar in basic mechanical design as defined in 4.5.2.1.2, subgroup 2 inspection may be omitted for that 12-month period.

4.5.2.1.2 Similar delay lines. A similar delay line (see 6.4) shall meet the following criteria. Similarity may be established when the Commander or Agency concerned is satisfied that the unit compares favorably with the tested and approved unit as follows:

(a) Same general environment.
(b) Same external and internal mountings, similar shape, same case construction, nominal wall thickness within 25 percent when a case is used, linear case dimensions not greater than 150 percent nor less than 70 percent of the corresponding dimensions, total volume of case not greater than 250 percent. (For uncased units, the envelope dimensions shall be considered as the case dimensions.)
(c) Same terminal construction and material, including insulating and gasketing parts, same or lower terminal-strength requirement for the same size of terminals, same or lower dielectric withstanding voltage requirement for the same size of terminals, same or lower current rating for corresponding terminals and the case wall or other grounded surfaces.
(d) Same or greater wire size for corresponding windings.
(e) Same processing and material for case, finish, marking, insulation, and filling.
(f) Same or lower figure or merit.
(g) Capacitors shall have the same dielectric material, same or higher voltage rating, and same or lower applied voltage stress.

4.5.2.1.3 Failures. If one or more sample units fail to pass group C inspection, the sample shall be considered to have failed.

4.5.2.1.4 Disposition of sample units. Sample units which have been subjected to group C inspection shall not be delivered on the contract or purchase order.

4.5.2.1.5 Noncompliance. If a sample fails to pass group C inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group C inspection shall be repeated on additional sample units, (all inspection, or the inspection which the original sample failed, at the option of the Government). Groups A and B inspections may be reinstituted; however, final acceptance shall be withheld until group C reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity.

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4.5.3 Packaging inspection. The sampling and the inspection of the preservation-packaging, packing and container marking shall be in accordance with the requirements of MIL-E-17555.

4.6 Methods of examination and test.

4.6.1 Solderability (see 3.6). Delay lines shall be tested in accordance with 4.6.1.1 or 4.6.1.2 as applicable (see 3.1). The method in 4.6.1.1 is preferred and shall be specified whenever practicable

4.6.1.1 Solder bath method. Delay lines shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

(a) Special preparation of specimen. - Sample units shall not have been soldered during any of the previous tests.
(b) Number of terminations of each part to be tested - A minimum of two of each type of terminal.

4.6.1.2 Soldering iron method. The test shall be performed on solder terminations, attached to the delay lines. The solder shall conform to type S, composition Sn60, of QQ-S-571. The flux shall conform to type A or W (as applicable) of MIL-F-14256. The temperature of the bit shall be 300 to 350°C. The iron and solder shall be applied to the termination for 10 seconds. The solder shall be applied for the first 2 seconds. Tinning, as evidenced by the free flowing of the solder with proper wetting of the termination, shall be completed within the first 2 seconds. The delay line under test shall remain under room ambient conditions for recovery for 15 minutes, before final examination is made.

(a) Special preparation of specimen - The surface shall be smooth and properly tinned and the solder terminations shall not have been soldered during any previous test.
(b) Number of terminations - In accordance with 4.6.1.1.
(c) Examinations of terminations - In accordance with method 208 of MIL-STD-202.
(d) Soldering irons - The soldering iron shall have one of the following bit sizes:
    1. 0.3 inch diameter, 1.25 inch exposed length reduced to a wedge shape, over a length of approximately 0.4 inch.
    2. 0.125 inch diameter, 0.5 inch exposed length reduced to a wedge shape, over a length of approximately 0.2 inch.
(e) Point of application of soldering iron - 1/4 inch from the nearest insulating material or to one-half the exposed length of the terminal, whichever point is closer to the insulating material.
(f) Examination after test - Delay lines shall be examined as specified in method 208 of MIL-STD-202.

4.6.2 Resistance to solvents (see 3.7). Delay lines shall be tested in accordance with method 215 of MIL-STD-202. After the final immersion and brushing, delay lines shall be allowed to dry. When dry, the delay lines shall be examined under a four-power magnification.

4.6.3 Visual and mechanical examination. Delay lines shall be examined to verify that the materials, design, construction, terminals, physical dimensions, interchangeability, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4, 3.5, 3.24, and 3.25).

4.6.4 Resistance to soldering heat (see 3.8). Delay lines shall be tested in accordance with 4.6.4.1 or 4.6.4.2, as applicable (see 3.1). The method in 4.6.4.1 preferred and shall be specified whenever practicable.

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4.6.4.1 Solder bath method. Delay lines shall be tested in accordance with method 210 of MIL-STD-202. The following details shall apply:

(a) Special preparation of specimen - Sample units shall not have been soldered during any of the previous tests.
(b) Depth of immersion in the molten solder - To a point 1/4 inch from the nearest insulating material to one-half the exposed length of the terminal, whichever point is closer to the insulating material.
(c) Test condition - A.
(d) Examination after test - The delay line shall be visually examined and there shall be no seepage of the impregnant, loosening of the terminals, or other mechanical damage. The windings of delay lines shall be checked for continuity.

4.6.4.2 Soldering iron method. The test shall be the same method specified in 4.6.1.2 except all solder terminations shall be tested, and the delay lines shall be examined after the test as specified in 4.6.4.1(d).

4.6.5 Terminal strength (see 3.9). Delay lines shall be tested as specified in 4.6.5.1 to 4.6.5.2 inclusive, as applicable. After each test, the terminals shall be examined for loosening, rupturing, and other mechanical damage. Unless otherwise specified (see 3.1), all terminals on each test sample shall be subjected to the applicable tests, up to a maximum of four identical terminals per sample.

4.6.5.1 Pull.

4.6.5.1.1 Solid-wire lead terminals (other than printed-circuit terminals). Delay lines shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.
(b) Points of measurement - A force shall be applied in the direction of the axis of termination and shall be increased gradually until the magnitude specified in table V is reached, and shall be held at the value for at least 10 seconds.

4.6.5.1.2 Solder terminals. Delay lines shall be tested in accordance with method 211 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.
(b) Points of measurement - A force as specified in table V shall be applied to each terminal at the point where the lead from the external circuit connects to it. The force shall be applied in any direction, including the weakest, and shall be increased gradually to the specified magnitude and shall be held at the value for at least 5 to 10 seconds.

TABLE V. Pull.

Cross-sectional area of electrode at its smallest point at which lead from external circuit connects	Force
Circular mils	Pounds
≤ 2,000	2.5
> 2,000	5.0

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4.6.5.1.3 Printed-circuit and pin type terminals. Delay lines shall be tested in accordance with method 211 of MIL-STD-202. The following detail and exception shall apply:

(a) Test condition - A.
(b) Applied force - A force of 2.5 pounds shall be gradually applied to each terminal in the direction of the axis of the terminal as shown on figure 2.

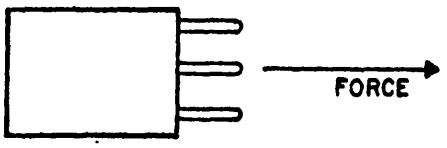


FIGURE 2. Direction of force to be applied to each pin on unit.

4.6.5.2 Twist or bend.

4.6.5.2.1 Solid-wire lead terminals (other than printed circuit terminals). Following the test specified in 4.6.5.1.1, delay lines shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test condition - D.
(b) Application of torsion - The body of the part or the clamped terminal shall be rotated through 360 degrees about the original axis of the bent terminal, in alternating directions, for a total of five rotations, at the rate of approximately 3 seconds per rotation.

4.6.5.2.2 Flat solder terminals. Any terminal that shows permanent deformation greater than 15 degrees of the metal portion of the terminal in the terminal-pull test specified in 4.6.5.1.2 shall be tested in accordance with method 211 of MIL-STD-202. This test does not apply to terminals which show permanent deformation, but are not designed to be bent 45 degrees. The following details and exceptions shall apply:

(a) Test condition - B.
(b) Number of bending operations - Five times through an angle of 90 degrees (45 degrees each side of center).

4.6.6 Seal (see 3.10). Delay lines shall be tested in accordance with 4.6.6.1 and 4.6.6.2, when applicable. When specified (see 3.1), the alternate test specified in 4.6.6.2 shall be used in lieu of the test specified in 4.6.6.1. For quality conformance inspection, any delay line which shows evidence of leakage may be given remedial treatment, provided that evidence is submitted to show that such remedial treatment is adequate and that all of the other delay lines in the lot have been given similar remedial treatment where required.

4.6.6.1 All delay lines. Delay lines shall be immersed for 2 to 3 minutes in a bath of water, or any other suitable liquid of no greater density and surface tension, maintained at a temperature of at least 85°C. The temperature of the delay line shall not exceed 40° C at the time of immersion.

4.6.6.2 Alternate test. When specified (see 3.1), the delay lines shall be immersed in a container of water containing approximately one percent aerosol which shall then be placed in a vacuum chamber. The pressure shall be reduced to 3.4 inches of mercury and maintained at this value for a period of at least 3 minutes.

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4.6.7 Dielectric withstanding voltage (see 3.11). Delay lines shall be tested in accordance with 4.6.7.1, 4.6.7.2, and 4.6.7.3 (as applicable).

4.6.7.1 At atmospheric pressure. The delay lines shall be subjected to the dielectric withstanding voltage test outlined in method 301 of MIL-STD-202. Unless otherwise specified (see 3.1), the test voltage shall be 500 volts dc. The following details shall apply:

(a) Duration of application of test voltage - Not less than 1 second or more than 5 seconds. The duration of the test shall begin when 95 percent of the test potential is reached.
(b) Points of application of test voltage - Between mutually insulated terminals and between terminals and case, if applicable.
(c) Limiting value of surge current - Shall not exceed 5 milliamperes.

4.6.7.2 Barometric pressure (reduced) (when applicable). Delay lines designed for operation above 10,000 feet shall be tested as specified in 4.6.7.1 and in accordance with method 105 of MIL-STD-202. The following details shall apply:

(a) Test condition or altitude in feet if below 30,000 feet - As specified (see 3.1).
(b) Magnitude of test voltage - As specified (see 3.1).
(c) Examination during and after test - Delay lines shall be examined for evidence of arcing, flashover, breakdown of insulation, and damage.

4.6.7.3 At reduced voltage. Delay lines shall be subjected to the dielectric withstanding voltage tests specified in 4.6.7.1 and 4.6.7.2 (when applicable) except that test voltages shall be 90 percent of the test values specified (see 3.1), and shall be applied for a period of 5 seconds, or 75 percent of the test values specified (see 3.1), applied for a period of 1 minute.

4.6.8 Insulation resistance (see 3.12). The delay lines shall be subjected to the insulation resistance test outlined in method 302 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.
(b) Measurements shall be taken between all taps and terminations wired together, and the case.

4.6.9 Electrical characteristics (see 3.1 and 3.13).

4.6 9.1 Pulse methods. The delay time, rise time, attenuation, distortion, and thermal stability test shall be conducted by applying an input pulse with a rise time 15 percent to 20 percent of the indicated output rise time to the delay line. The terminating impedance of the delay line shall be resistive and within ±1 percent of the value specified. Unless otherwise specified, the source impedance shall be within ±5 percent of the value of the terminating impedance. The method employed shall be acceptable to the Government.

4.6.9.1.1 Delay time. The time delay of pulses taken at each tap and the output of delay lines shall be measured to determine conformance with 3.13.1. In the case of sectionalized lines, the time delay of any combination of tandem sections shall be measured.

4.6.9.1.2 Rise time. The rise time of pulses taken at the output terminals (for sectionalized lines with all sections in tandem), shall be measured to determine conformance with 3.13.2.

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4.6.9.1.3 Attenuation. The amplitude of pulses taken at the input and output terminals (for sectionalized lines with all sections in tandem), shall be measured and the attenuation computed (see 3.13.3).

4.6.9.1.4 Distortion. The distortion of pulses taken at the output terminals (for sectionalized lines with all sections in tandem), when the delay line is fed with the specified input pulses shall be measured to determine conformance with 3.13.4.

4.6.9.1.5 Delay time variation with temperature. The delay line shall be enclosed in a suitable thermal chamber in which the temperature shall be variable over the applicable temperature range (see 3.13.5). The delay line shall be allowed to reach thermal stability at a sufficient number of temperatures covering the applicable temperature range (see 3.13.5) and at each stabilized temperature the delay time shall be measured and the delay time variation from the reference temperature delay time shall be recorded (see 3.13.5).

4.6.9.1.6 Thermal-age stability. After completion of the environmental test (except flammability), the change in the delay time, rise time, attenuation, and distortion shall be recorded (see 3.13.6).

4.6.10 Salt spray (corrosion) (when applicable, see 3.1 and 3.14). Delay lines shall be tested in accordance with method 101 of MIL-STD-202. Test condition shall be as specified (see 3.1).

4.6.11 Vibration (see 3.15). Delay lines shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Method of mounting - Delay lines shall be rigidly mounted by their normal mounting means.
(b) Test condition - As specified (see 3.1).
(c) Measurements - As specified (see 3.1).

4.6.12 Shock (see 3.16). Unless otherwise specified (see 3.1), delay lines shall be tested in accordance with method 213 of MIL-STD-202, test condition I.

4.6.13 Winding continuity (see 3.17). All windings of delay lines shall be tested for electrical continuity by any suitable means.

4.6.14 Thermal shock (see 3.18). Unless otherwise specified (see 3.1), delay lines shall be tested in accordance with method 107 of MIL-STD-202, test condition A.

4.6.15 Immersion (see 3.19). Delay lines shall be tested in accordance with method 104 of MIL-STD-202. The following details shall apply:

(a) Test condition - A.
(b) Measurements after final cycle - As specified (see 3.1).

4.6.16 Moisture resistance (see 3.20). Unless otherwise specified (see 3.1), delay lines shall be tested in accordance with method 106 of MIL-STD-202.

4.6.17 Flammability (when specified) (see 3.21). Unless otherwise specified (see 3.1, delay lines shall be tested in accordance with method 111 of MIL-STD-202.

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4.6.18 Life (when specified, see 3.1 and 6.2.2)(see 3.22). Delay lines shall be tested in accordance with method 108 of MIL-STD-202. The following details shall apply:

(a) Distance of temperature measurements from specimens - 6 inches.
(b) Test temperature and tolerance - As specified (see 3.1 and 6.2.2).
(c) Operating conditions - As specified (see 3.1 and 6.2.2).
(d) Test condition - D.
(e) Measurements - As specified (see 3.1 and 6.2.2).

4.6.19 Fungus (see 3.23). Unless certification is provided, delay lines shall be tested in accordance with method 508 of MIL-STD-810.

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified (see 6.2), the requirements for packaging shall be in accordance with MIL-E-17555.

5.2 Navy procurements. The preparation for delivery requirements specified on Navy contracts apply only for direct Government shipments. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by Navy contractors or their authorized distributors shall be specified in the individual orders.

6. NOTES

6.1 Intended use. Delay lines covered by this specification are used in electronic equipment where a pulse delay is required.

6.2 Ordering data.

6.2.1 For delay lines covered by coordinated specification sheets, procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number and date of the applicable specification sheet and the complete military part number.
(c) Packaging of delay lines, if other than that specified (see 5.1).
(d) Whether first article inspection is to be performed (see 3.2, 4.4, and 6.3).

6.3 First article. When a first article is required for inspection and approval (see 3.2, 4.4, and 6.2).

6.4 Similar delay lines. A similar delay line is defined as a delay line which, when compared to the first article approved delay line, meets the criteria specified in 4.5.2.1.2.

6.5 Figure of merit. The figure of merit is a measure of the minimum inherent delay to rise-time ratio that shall be expected of the line.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Navy - SH, AS, OS
Air Force - 17

User activities:
Army - MI
Navy - MC

Preparing activity:
Navy - EC

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